

Research Paper Template: What Goes in Each Section

Use this template to see what belongs in each part of a research paper. The colors show the job of each idea. The running example is ECG image digitization: turn a photo or scan of a paper ECG into a clean digital signal.

How to read the colors

- **Problem** red marks the problem you solve.
- **Method** teal marks the method or idea.
- **Result** green marks the result or claim.
- **Gap** orange marks the gap in past work.

The whole paper at a glance

Section	Its one job
1. Title	Name the work in one clear line
2. Abstract	The whole paper in one short paragraph
3. Introduction	Walk from the big picture to your question
4. Related Work	Show past work and where you fit
5. Methodology	Explain your method so others can rebuild it
6. Experiments	Describe the test setup
7. Results	Report what the numbers say
8. Discussion	Explain what the results mean
9. Limitations	Say what your work does not cover
10. Conclusion	Wrap up and point forward
11. References	List every source you cited

1 Title

Purpose: name the paper so a reader knows the topic in one line.

What to put here

- The **method or idea** and the **problem** it solves.
- Keep it short, plain, and specific. No hype words.

ECG example

Robust ECG Image Digitization Under Common Image Damage: A Method by Method Benchmark on PhysioNet 2024

2 Abstract

Purpose: give the whole paper in one short paragraph.

What to put here

- One or two lines of context, then the **problem**.

- What you did (the **method**) and the main **result**.
- Aim for 4 to 6 short sentences. No citations, no tables.

ECG example

Paper ECG scans often carry creases, shadows, and ink smudges. These flaws can lower the quality of the digital signal. Past work reports one overall score and does not show how each type of damage changes the result. We test three digitization methods on PhysioNet 2024 and group the images by damage type. We find that the best method changes with the damage type, and no single method wins on every case.

3 Introduction

Purpose: walk the reader from the big picture down to your exact question.

What to put here

An introduction is built in a clear order. Follow these steps.

- **Step 1. Wide context.** Say why the area matters.
ECG: many clinics still keep heart records only on paper.
- **Step 2. The problem.** Name the specific pain.
ECG: scanned paper ECGs are often damaged by creases, shadows, and smudges.
- **Step 3. What is missing.** Say what current work does not do.
ECG: current tools report a single overall accuracy score.
- **Step 4. The gap.** Name the gap in one clear line.
ECG: no study shows if accuracy holds up for each separate type of damage.
- **Step 5. The question.** Turn the gap into a question.
ECG: does digitization accuracy hold up for each type of image damage, method by method?
- **Step 6. The contributions.** List what you add.
 - We benchmark three digitization methods, one damage type at a time.
 - We report SNR in dB for each method and each damage type.
 - We show which method to trust for which kind of damage.

4 Related Work

Purpose: show what others did and where your work sits.

What to put here

- Group the prior **methods** and say what each one does.
- Give the limit of each group, then point to the **gap**.
- Compare the works. Do not just list them one by one.

ECG example

Early methods trace the ECG line with hand written rules. Newer methods use deep networks to read the signal. Both report one overall score on clean or lightly damaged images. None breaks the score down by damage type, which is the gap we fill.

5 Methodology

Purpose: explain your method so another person could rebuild it.

What to put here

- The **method** step by step, with inputs and outputs.
- The model or pipeline, and the settings you used.
- Enough detail that a reader could repeat your work.

ECG example

Our pipeline has five steps: load the ECG image, remove the printed grid, find the signal line, map the pixels to millivolts, and save the digital signal. We run this same pipeline for each of the three methods so the comparison is fair.

6 Experiments

Purpose: describe the setup you used to test the method.

What to put here

- The dataset, the benchmark, and the **metric** you measure.
- The baselines you compare against.
- How you split the data. This is the plan, not the numbers yet.

ECG example

We use the PhysioNet 2024 benchmark. The metric is **SNR in dB**, where higher is better. We compare three methods, called A, B, and C. We split the images into five damage groups: clean scan, creases, shadow, ink smudge, and low resolution.

7 Results

Purpose: report what the numbers say, with no opinion yet.

What to put here

- The **results** in clear tables, grouped the way you planned.
- A direct answer to your question, backed by the numbers.

ECG example

Table: example SNR in dB, higher is better. Numbers are illustrative.

Damage type	Method A	Method B	Method C
Clean scan	24.8	24.1	22.0
Creases	18.2	20.4	15.1
Shadow	16.7	19.9	14.0
Ink smudge	12.1	17.6	10.5
Low resolution	21.3	20.0	19.2

Method B gives the highest SNR on creases, shadow, and ink smudge. Method A wins on clean scans and low resolution.

8 Discussion

Purpose: explain what the results mean and why.

What to put here

- Read the **result** and say what it tells us.
- Link back to the **gap** and your question.
- Say when the method works well and when it does not.

ECG example

The **best method** depends on the type of damage. This **answers the gap directly**: one overall score hides these swings and can point a user to the wrong method for their damaged scans.

9 Limitations

Purpose: say honestly what your work does not cover.

What to put here

- Where the **method** breaks or was not tested.
- Limits of the data and the setup. Be specific and honest.

ECG example

We test five damage types on one benchmark only. **We do not test images that carry two kinds of damage at the same time**, so mixed damage is still an open case.

10 Conclusion

Purpose: wrap up in a few lines and point forward.

What to put here

- Restate the **problem** and the main **result** in short.
- One line on useful future work.

ECG example

Damaged ECG scans are hard to digitize. We showed that the **best method changes with the type of damage**. Future work should test mixed damage and add more methods.

11 References

Purpose: list every source you cited, in one steady style.

What to put here

- A full citation for each work: authors, title, venue, and year.
- Use the same style for all entries. List only sources you actually cited.
- Use the url command for web links so they stay clickable.

ECG example

- PhysioNet 2024 Challenge. Digitization of ECG images. 2024. <https://physionet.org>
- A. Author, B. Author. Deep ECG signal recovery from paper scans. Journal of Biomedical Signals, 2023.